Context Free Grammar

What we have learnt so far

- DFA / NFA computers with extremely limited memory.
- Regular Expressions
  Recipe to describe languages that are regular.

How about creating a slightly powerful computational model?

$$A = \{0^n 1^n \mid n \geq 0\}$$

Instead going directly to the new computational model, let's try to learn about something called context free grammar.

context free grammar to context free languages is what regular expressions are to regular languages.

☆ context free grammar is a collection of substitution rules also known as productions.

☆ Each rule appears as a line in the grammar.

☆ Each line comprises of a symbol, arrow, and a string.

terminals

Ex: $A \longrightarrow aAbB$

variable arrow variable

☆ The string can consist of variables, and other symbols called terminals.

☆ Variables are represented using capital letters, terminals are represented using lowercase letters, numbers or special symbols.

☆ One variable is designated as the starting variable.